MULTICORNER STA REPORT

16-Corner Post-Route Static Timing Analysis

VSDBABYSOC • SKY130HD • OpenROAD

ROLL NO: 24EG104A20

NAME: S N SRI TEJA

1. Objective

This report records a run of static timing analysis (STA) for the VSDBABYSOC design using the SKY130HD standard-cell library. The supplied terminal summary is used as the source for the numeric results; five supplied screenshots are included as implementation and run evidence.

2. Run setup

The Run reused the existing detailed-routed database (5_2_route.odb), global-route timing constraints (5_1_grt.sdc), and freshly extracted parasitics (5_2_route_fresh.spef). Each run loaded one standard-cell Liberty corner, together with the existing DAC and PLL macro libraries, and ran setup/hold path reports and slack summaries.

This is a repeat analysis on the same physical implementation, not a new placement-and-routing run. The DAC and PLL macro models were held fixed rather than independently swept across all 16 standard-cell corners.

3. Corner results

Values below are transcribed from the supplied Run summary and shown to its two-decimal-place precision. The duplicate entries ending in ".openroad" were excluded because they repeat the same corner results; only 16 unique corners are listed.

Corner	Worst setup slack (ns)	Worst hold slack (ns)	WNS max (ns)	TNS max (ns)
ff_100C_1v65	7.22	0.26	0.00	0.00
ff_100C_1v95	8.07	0.20	0.00	0.00
ff_n40C_1v56	5.99	0.30	0.00	0.00
ff_n40C_1v65	6.72	0.26	0.00	0.00
ff_n40C_1v76	7.36	0.23	0.00	0.00
ff_n40C_1v95	8.07	0.20	0.00	0.00
ss_100C_1v40	-3.55	0.92	-3.55	-390.07
ss_100C_1v60	1.24	0.65	0.00	0.00
ss_n40C_1v28	-29.21	1.80	-29.21	-14131.26
ss_n40C_1v35	-16.31	1.34	-16.31	-6593.49
ss_n40C_1v40	-10.87	1.12	-10.87	-3379.48
ss_n40C_1v44	-7.69	0.99	-7.69	-1900.72
ss_n40C_1v60	-0.45	0.67	-0.45	-5.89
ss_n40C_1v76	2.91	0.50	0.00	0.00
tt_025C_1v80	6.13	0.32	0.00	0.00
tt_100C_1v80	6.22	0.32	0.00	0.00

4. Interpretation

Six corners report negative worst setup slack: ss_100C_1v40, ss_n40C_1v28, ss_n40C_1v35, ss_n40C_1v40, ss_n40C_1v44, and ss_n40C_1v60. The most negative recorded setup slack is −29.21 ns at ss_n40C_1v28, with setup TNS of −14131.26 ns. These results indicate setup timing violations under the supplied constraints and timing models.

All 16 corners have positive reported worst hold slack, with the minimum reported hold slack of 0.20 ns. The summary reports hold TNS as 0.00 ns for all corners.

The summary marks each run PASS as a script/log completion status. It should not be interpreted as proof that timing passes in every corner: negative setup slack and negative setup TNS are present in several slow corners.

5. Limitations and physical-design notes

The STA results depend on the applied constraints and the available timing models. The earlier GUI timing views indicated unconstrained pins, so the report should not be treated as full sign-off without reviewing constraint coverage. Detailed routing previously reported metal-1 short violations; therefore, the implementation should not be described as DRC-clean.

6. Conclusion

The run reproduces the corner-wise timing pattern using the same routed implementation and parasitics. Typical and fast corners show positive setup slack, while six slow corners show setup violations. This Run is useful as a reproducibility check, but it does not constitute an independent physical implementation.

Figure 1. OpenROAD GUI view of the routed interconnect layout.

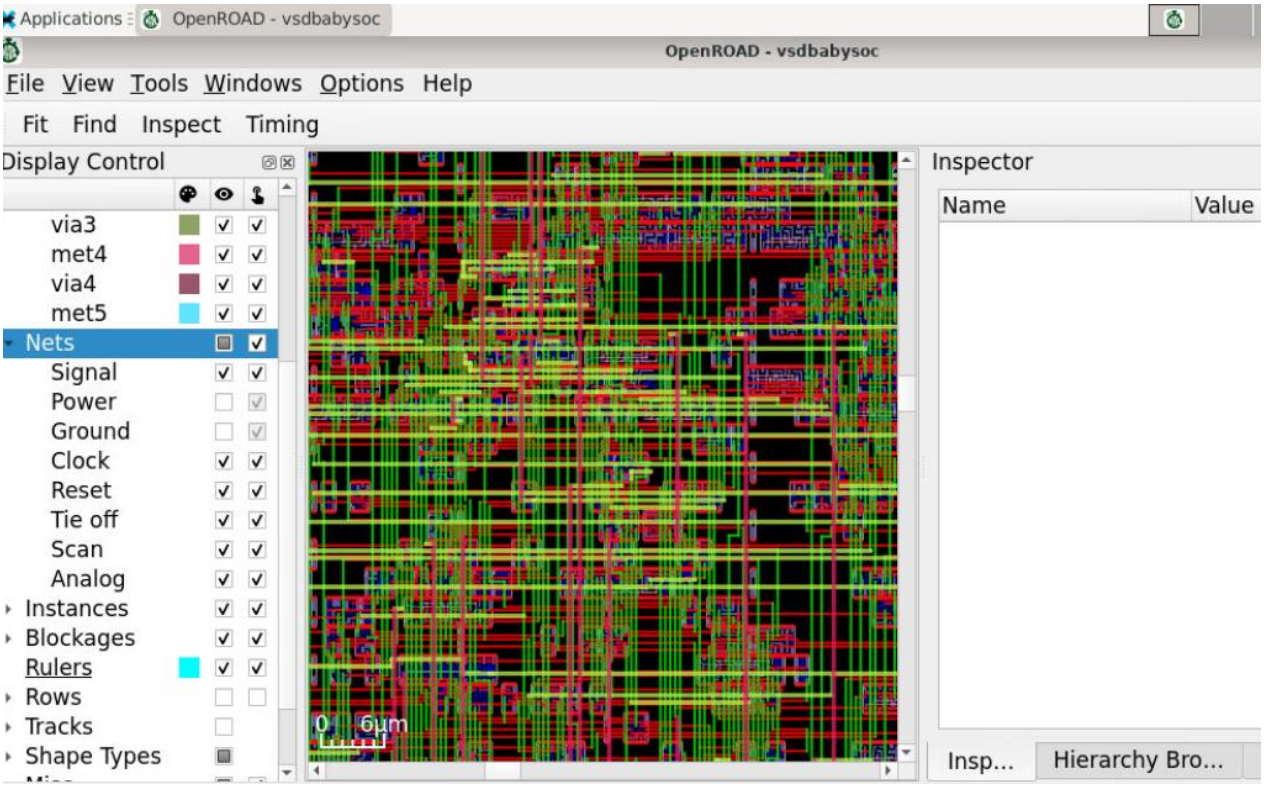


Figure 2. Zoomed routed-layout view showing interconnect detail.

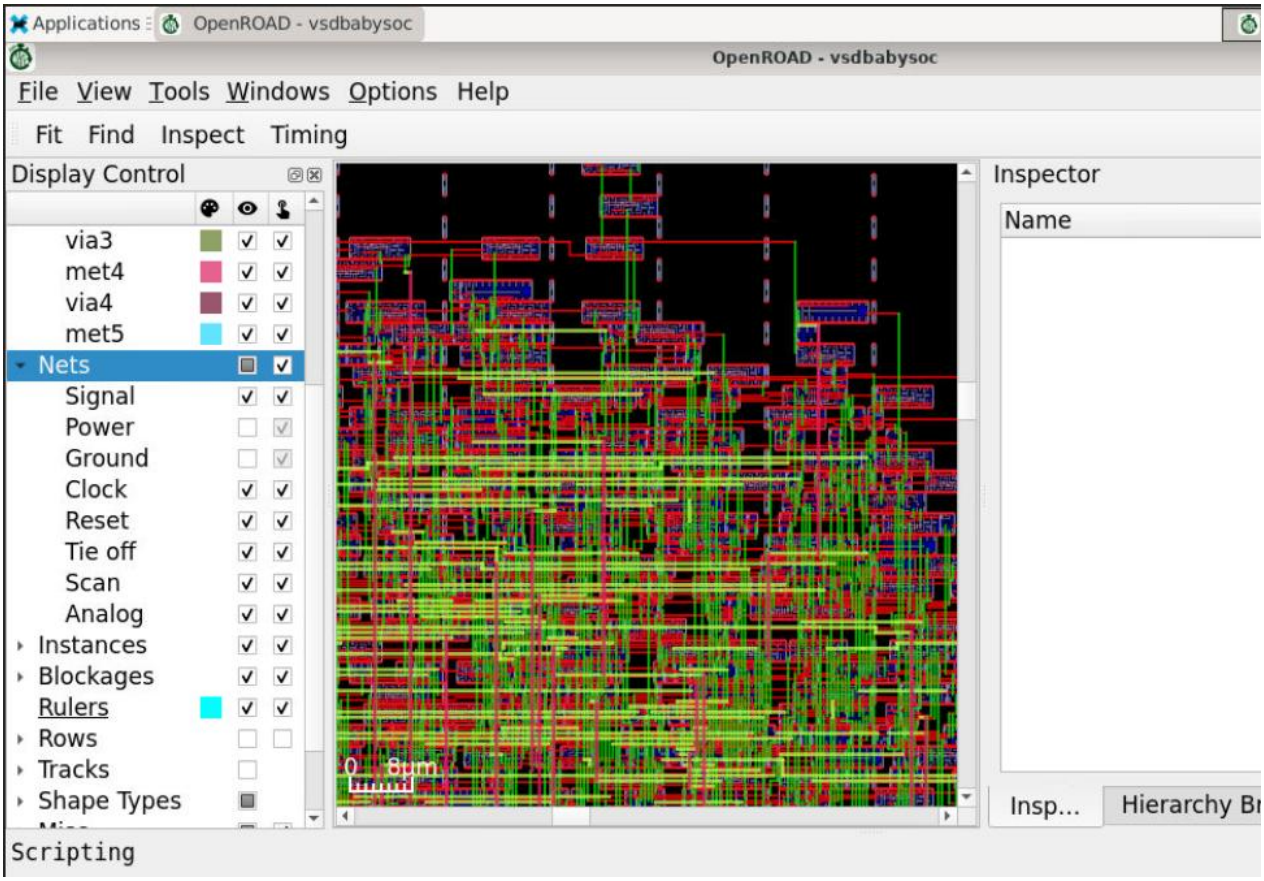


Figure 3. Detailed routing view around standard-cell instances.

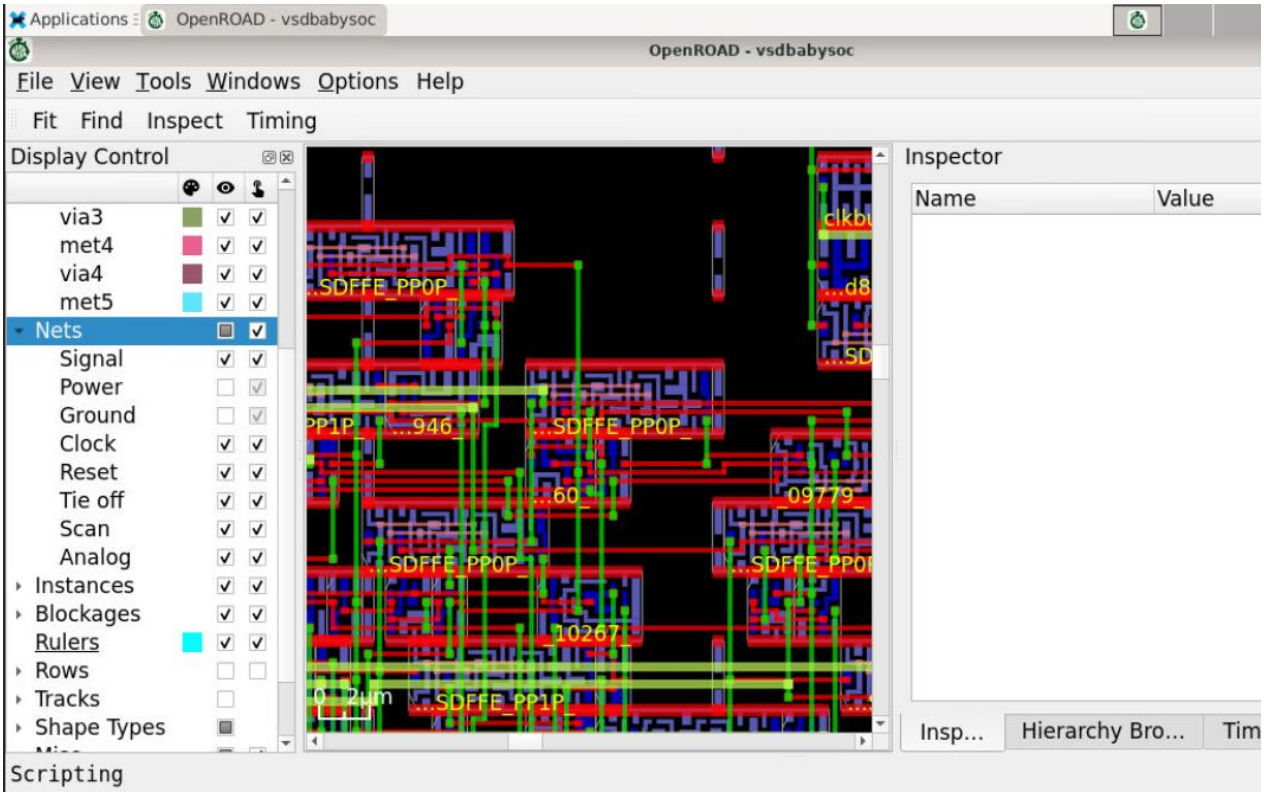


Figure 4. Terminal evidence from the 16-corner STA run.

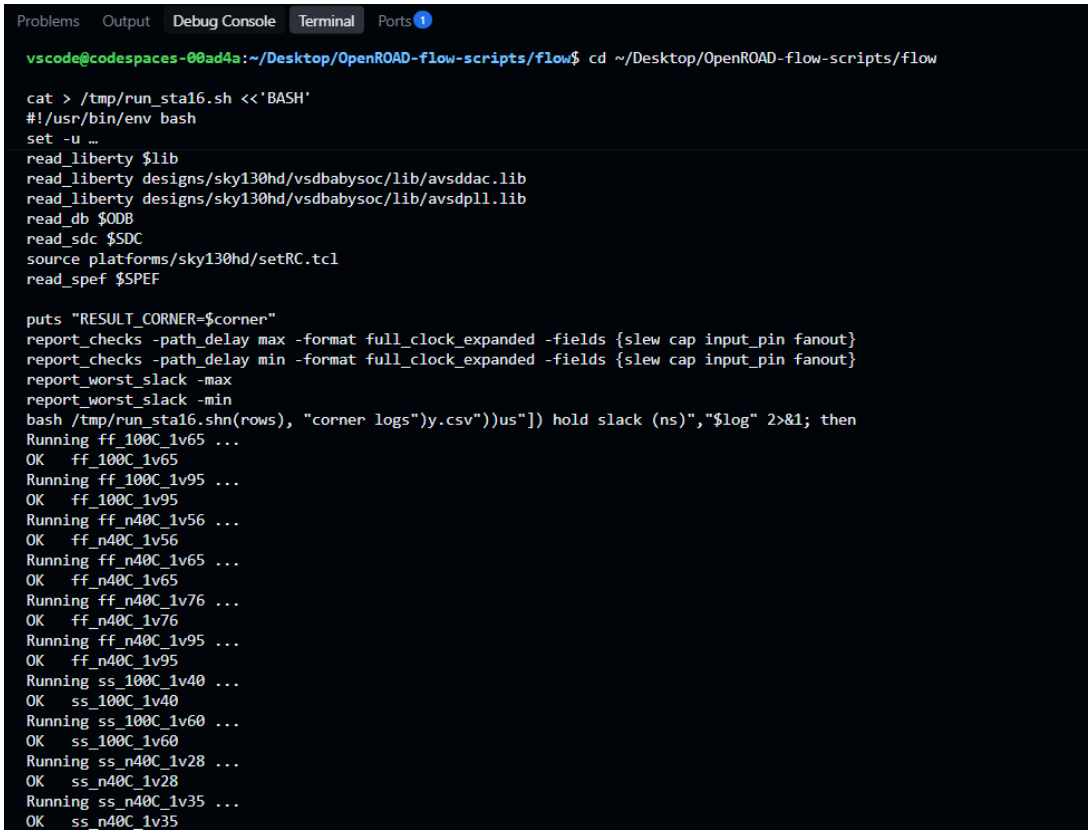


Figure 5. Terminal summary showing the unique corner results and duplicate log entries.

